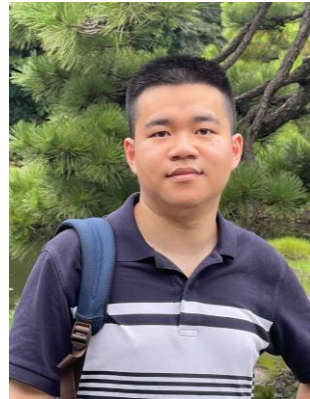


MulDar: Unleashing the Potential of Distributed COTS mmWave Radar by Exploiting Cross-Device Channels

Xinghua Sun, Qiancheng Li, Akshay Gadre (Presenter)

University of Washington



ELECTRICAL & COMPUTER
ENGINEERING

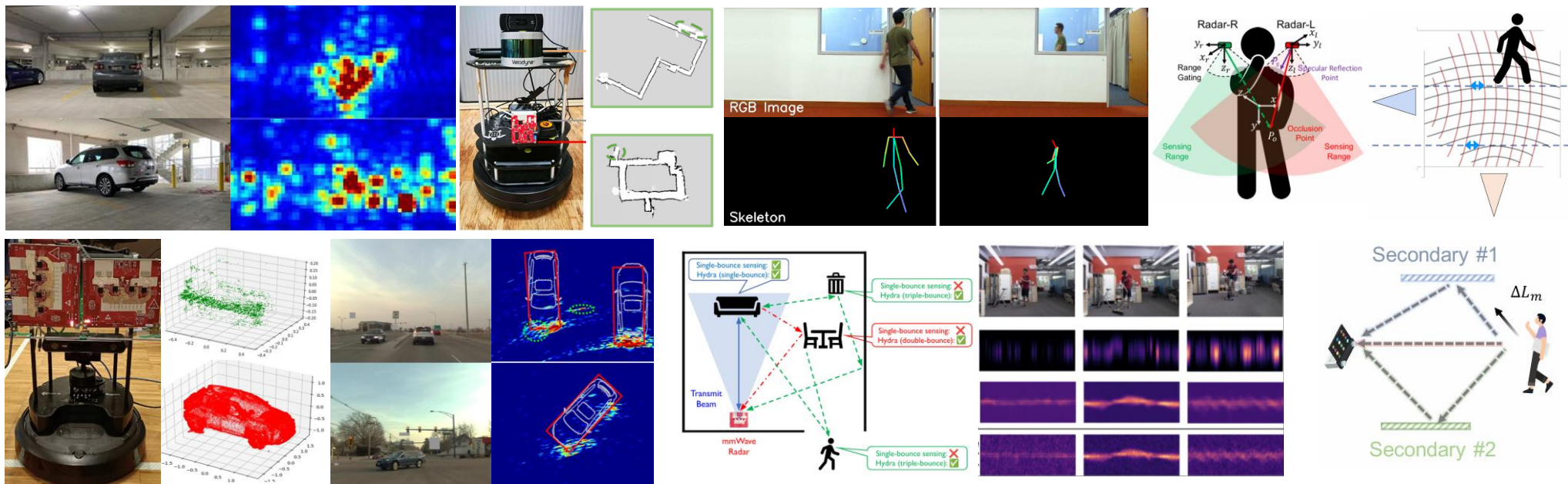
UNIVERSITY of WASHINGTON



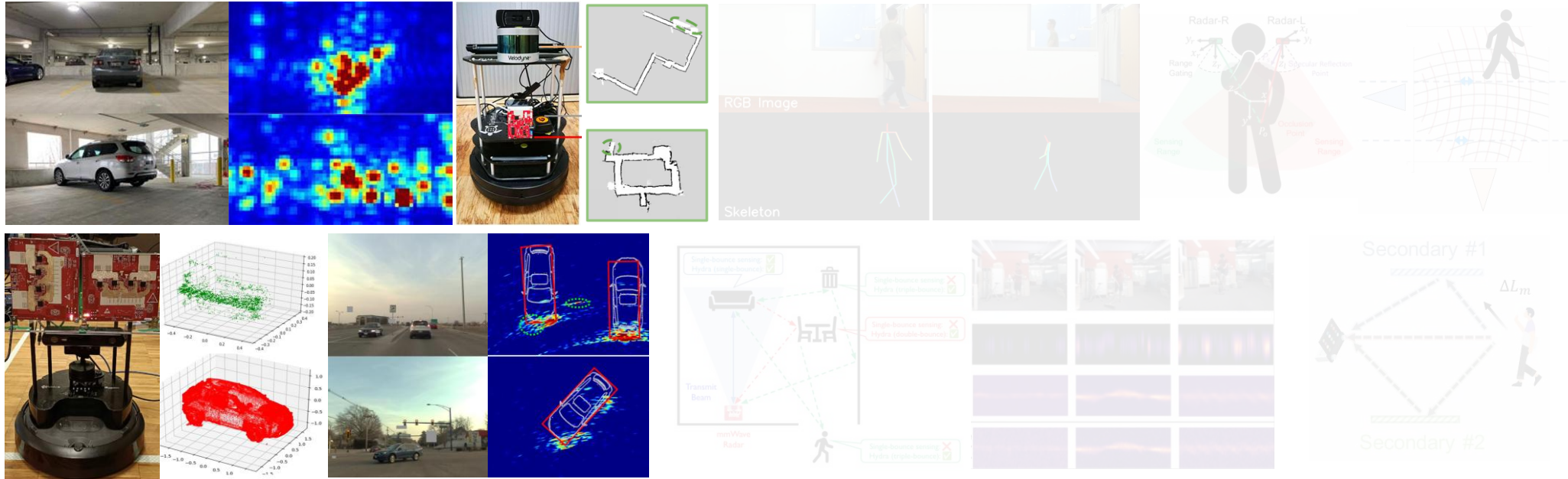
NEWT LAB
Networking and Emerging
Wireless Technologies Lab



Prior Work on Multi-view Radars

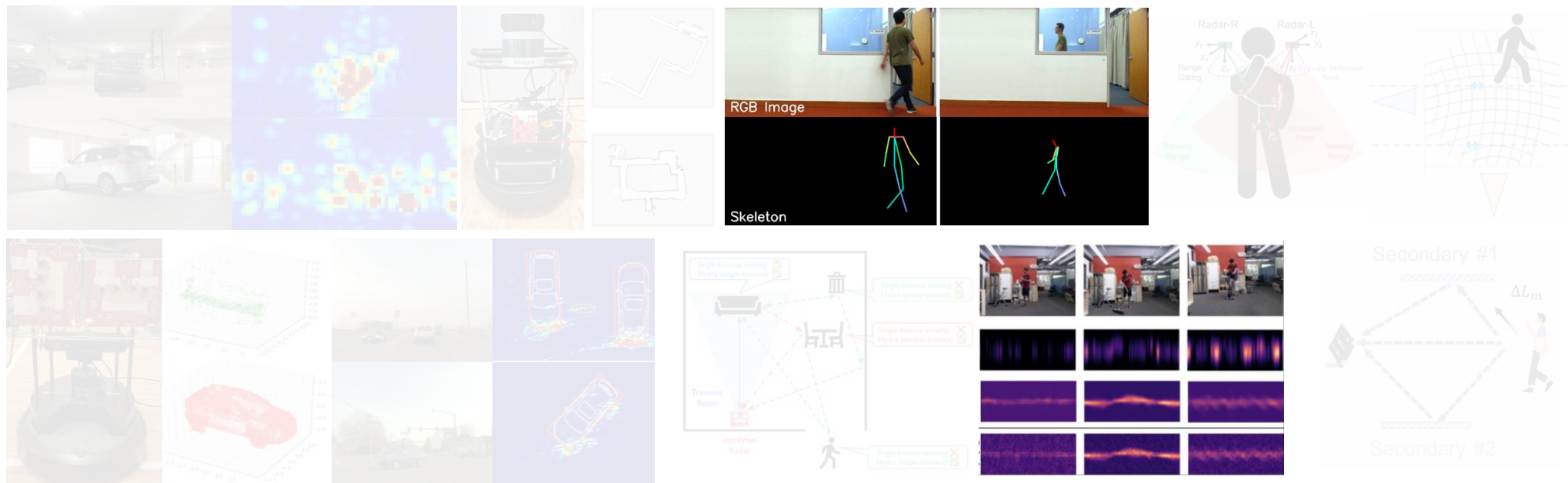


Prior Work on Multi-view Radars



3D Shape Reconstruction from Autonomous Driving Radars, BMVC '25
 Radatron: Accurate Detection Using Multi-Resolution Cascaded MIMO Radar, ECCV '22
 Through Fog High Resolution Imaging Using Millimeter Wave Radar, CVPR '20
 See through smoke: robust indoor mapping with low-cost mmWave radar, MobiSys '20

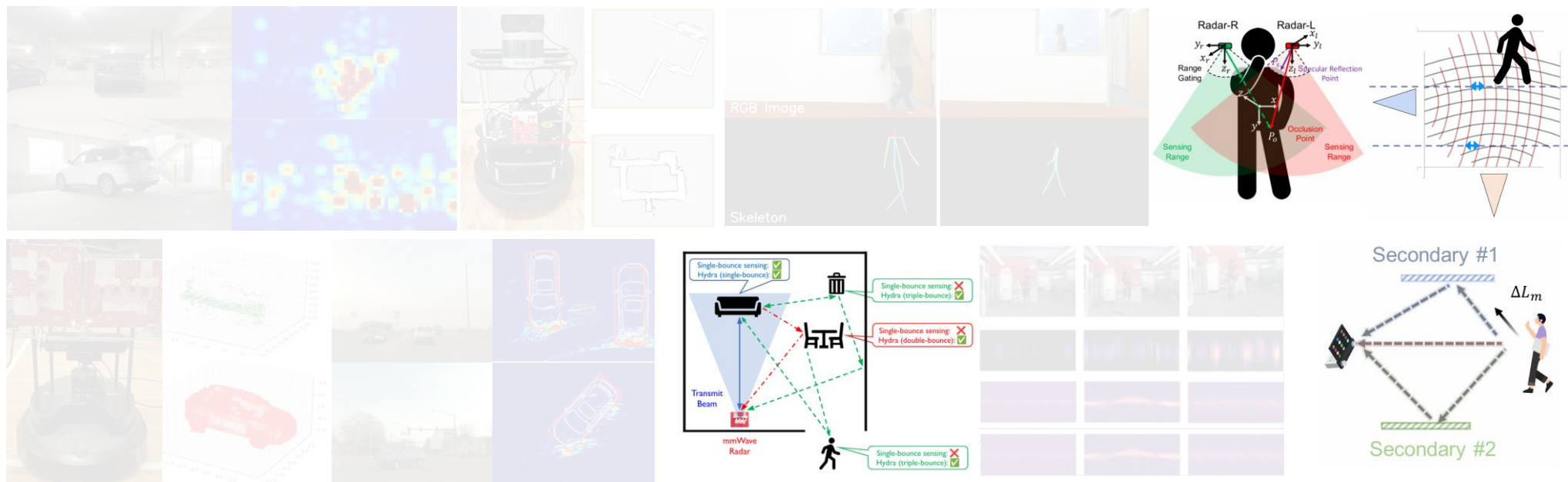
Prior Work on Multi-view Radars



Through-Wall Human Pose Estimation Using Radio Signals, CVPR '18

Vid2Doppler: Synthesizing Doppler Radar Data from Videos for Training Privacy-Preserving Activity Recognition, CHI '21

Prior Work on Multi-view Radars



SeRadar: Embracing Secondary Reflections for Human Sensing with mmWave Radar, MobiCom '25

Argus: Multi-View Egocentric Human Mesh Reconstruction Based on Stripped-Down Wearable mmWave Add-on, SenSys '25

MVDoppler-Pose: Multi-Modal Multi-View mmWave Sensing for Long-Distance Self-Occluded Human Walking Pose Estimation, CVPR '25

Hydra: Exploiting Multi-Bounce Scattering for Beyond-Field-of-View mmWave Radar, MobiCom '24

A Simple Experiment



Radar 1



Radar 2

A Simple Experiment



Radar 1

Radar 2

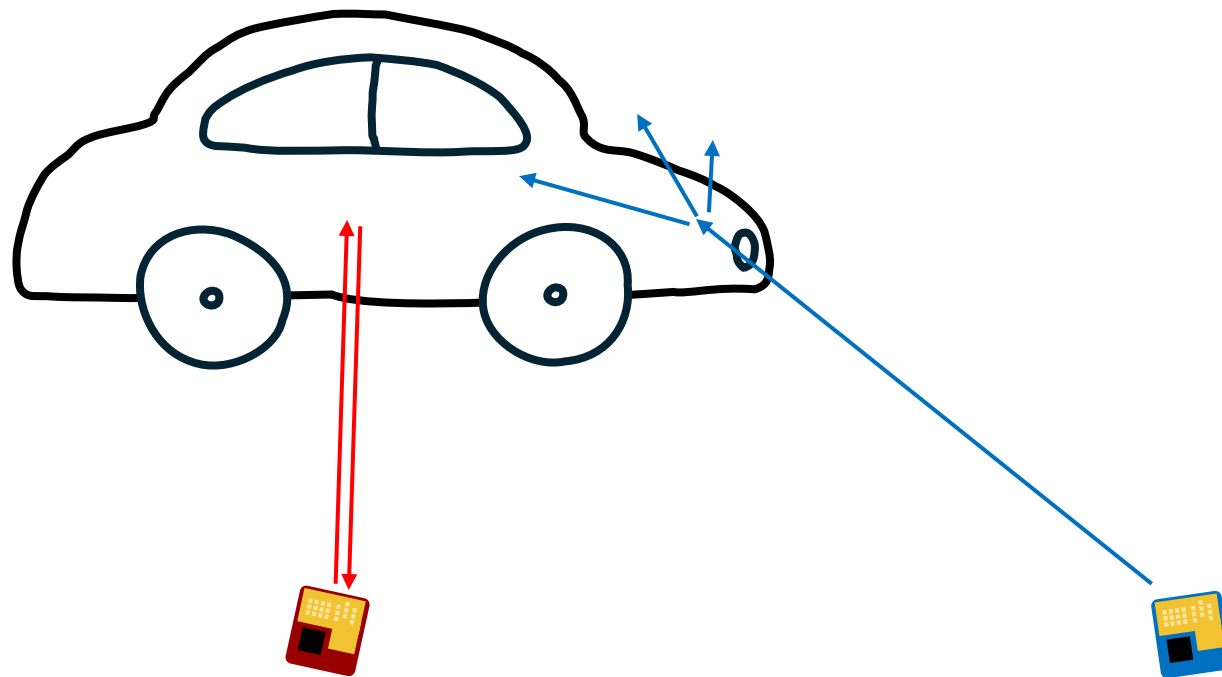


Combined Beamforming
Image of Radar 1 & 2

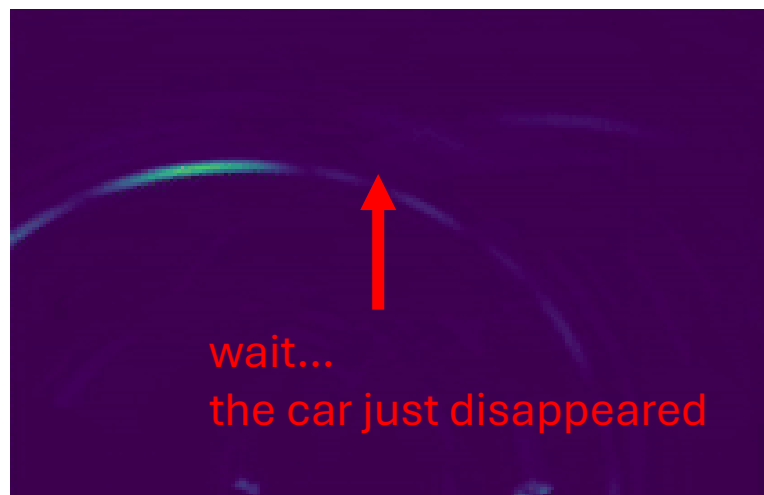
A Simple Experiment



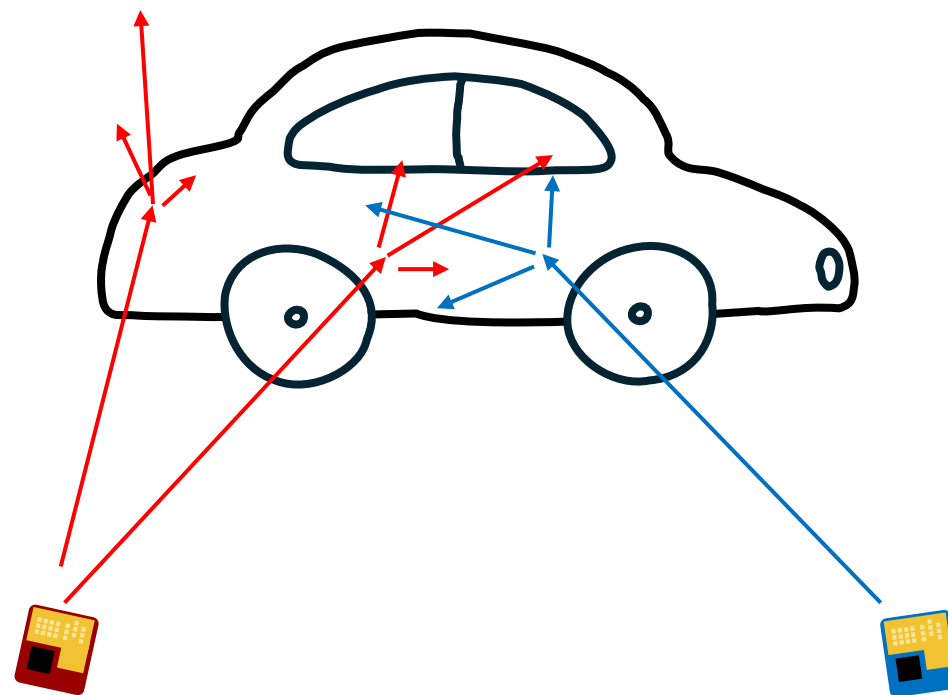
Combined Beamforming
Image of Radar 1 & 2



A Simple Experiment



Combined Beamforming
Image of Radar 1 & 2

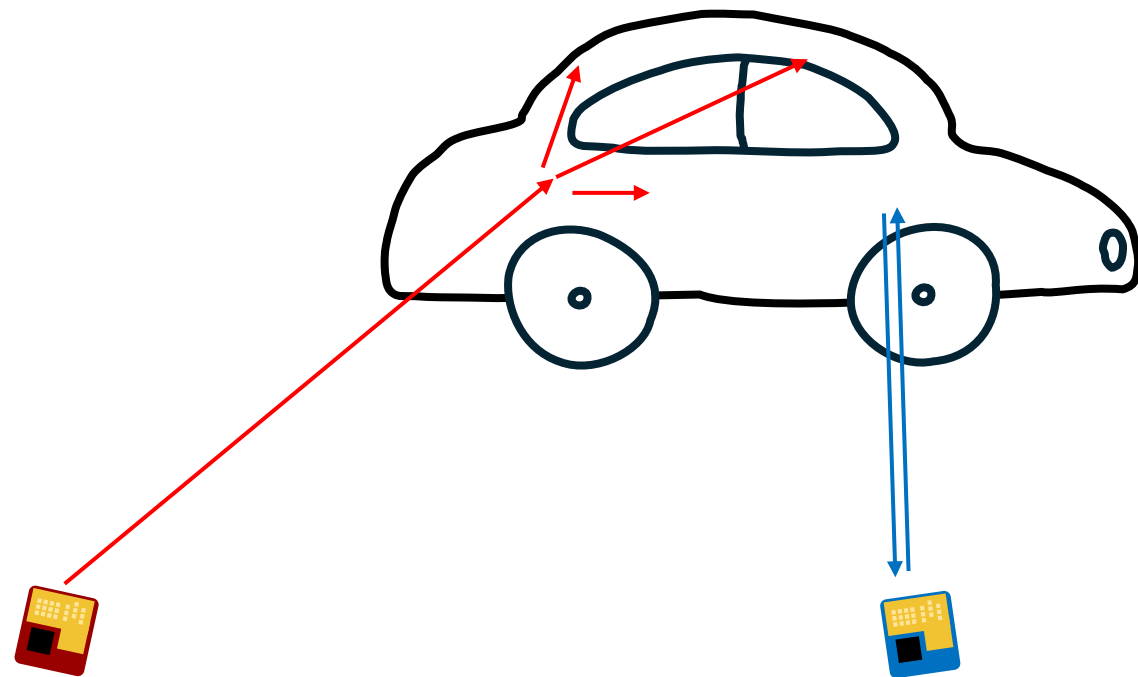


Specular Reflections:
the car is invisible to both radar!

A Simple Experiment



Combined Beamforming
Image of Radar 1 & 2

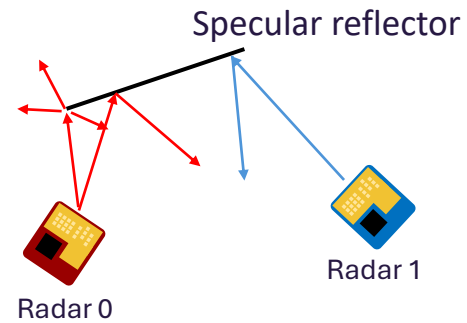


Fundamental Limitations of mmWave Radars

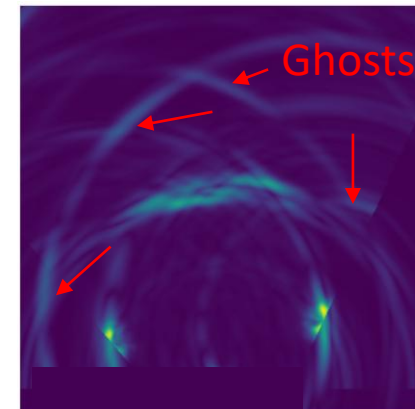
Sparse and noisy Information



Vague and incomplete shape

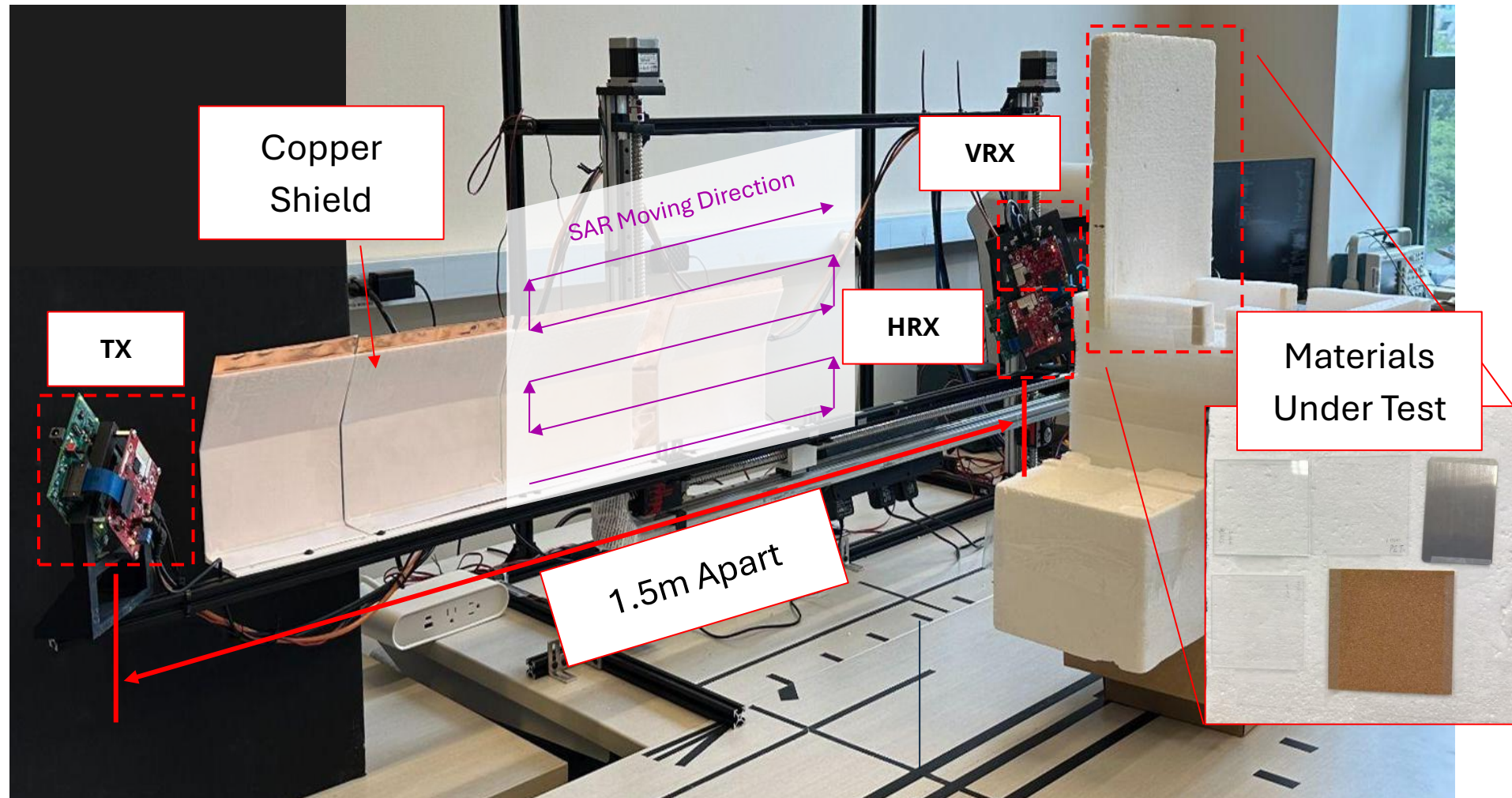


Invisible items

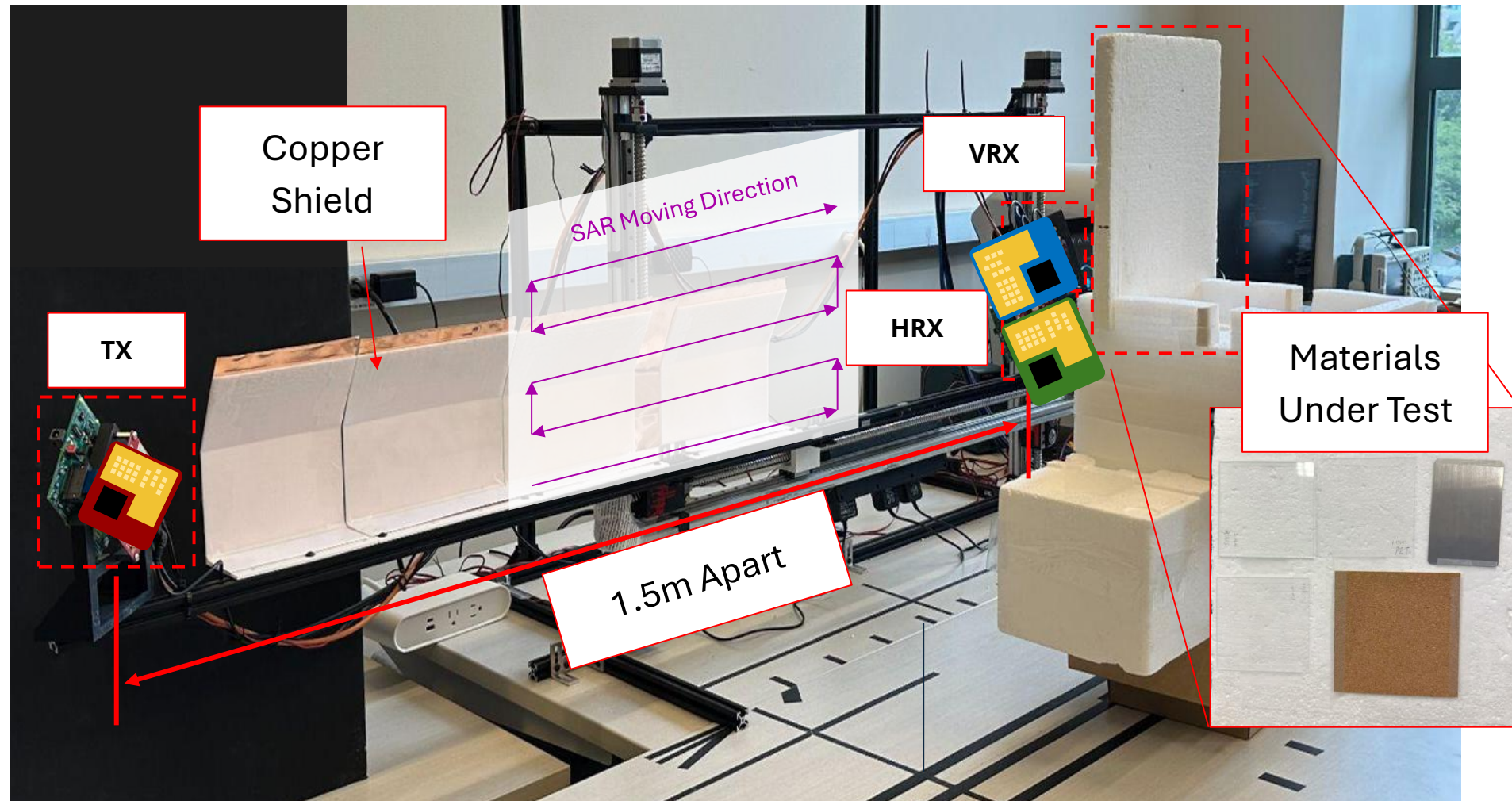


Ghost responses

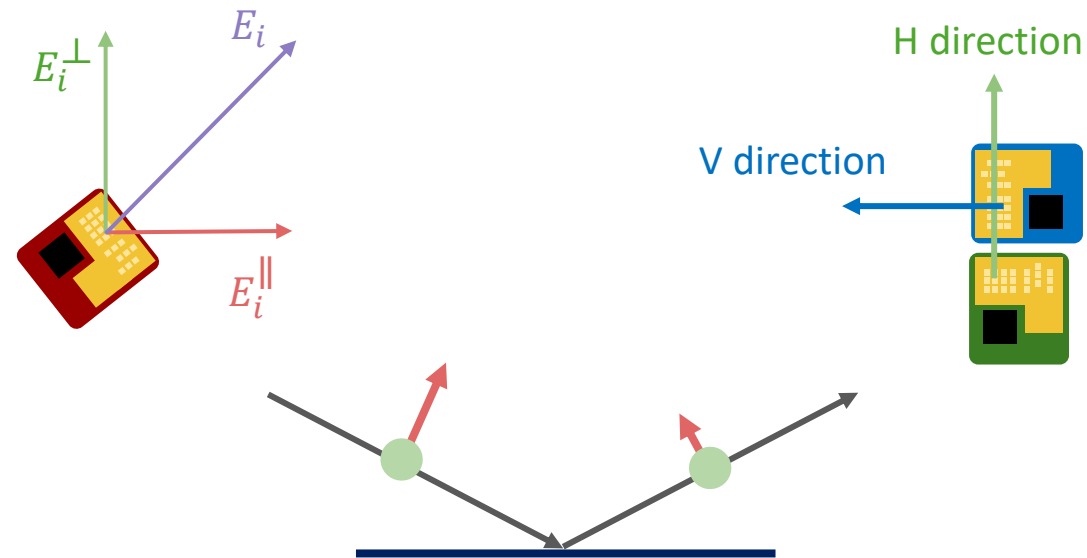
Our Prior Work on Distributed Radar: Bi-static Polarimetric SAR Imaging System



Our Prior Work on Distributed Radar: Bi-static Polarimetric SAR Imaging System

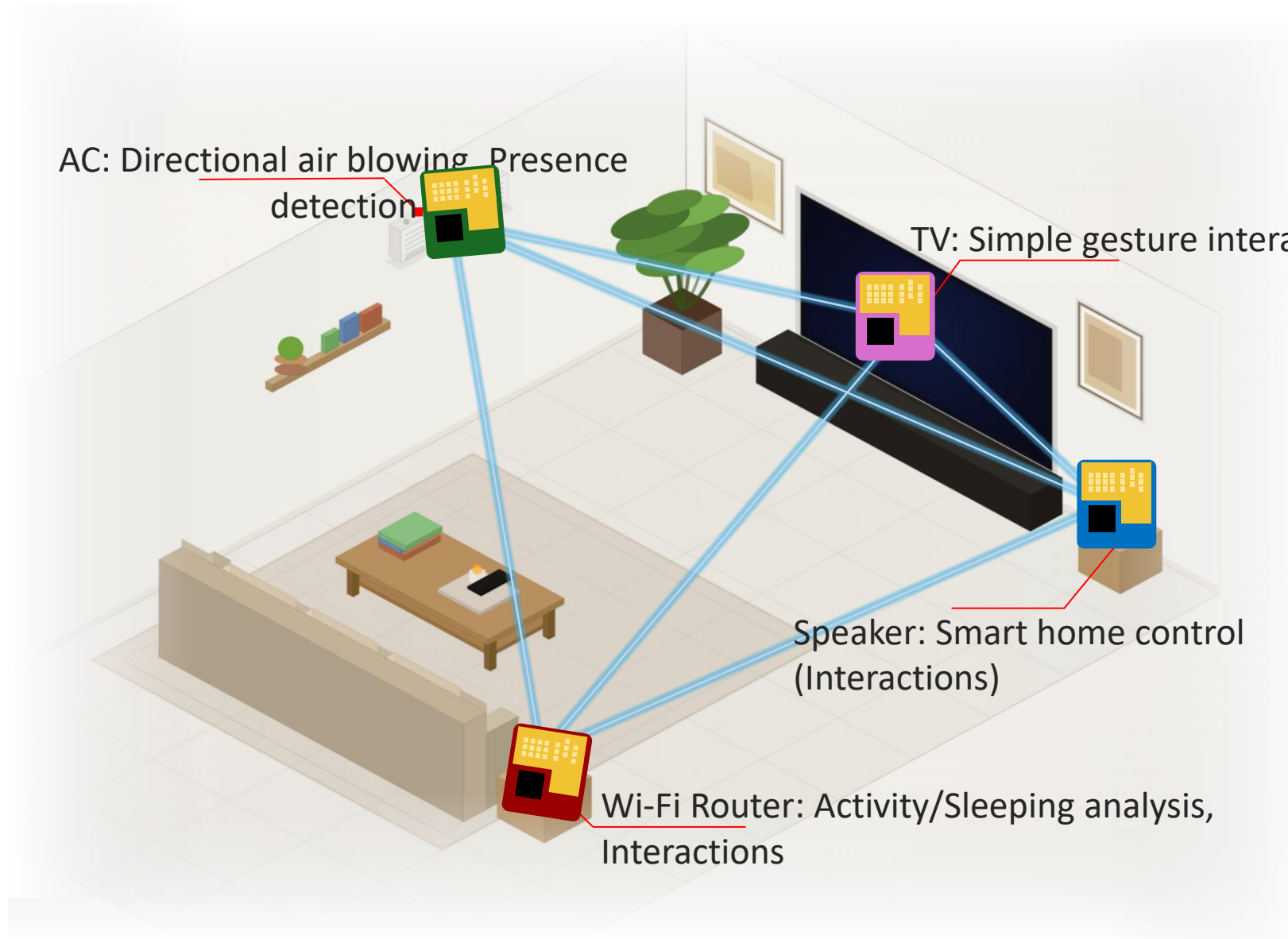


Distributed Setup: Spatial Channel Selection

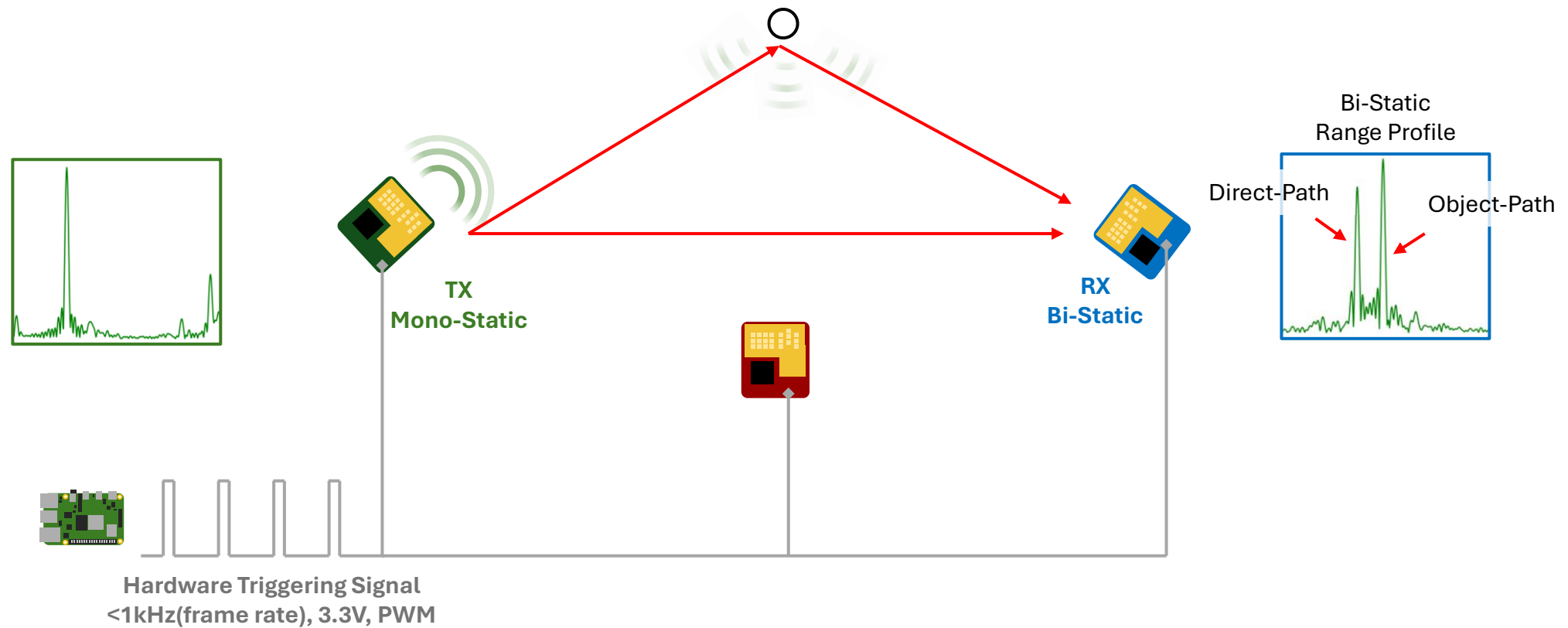


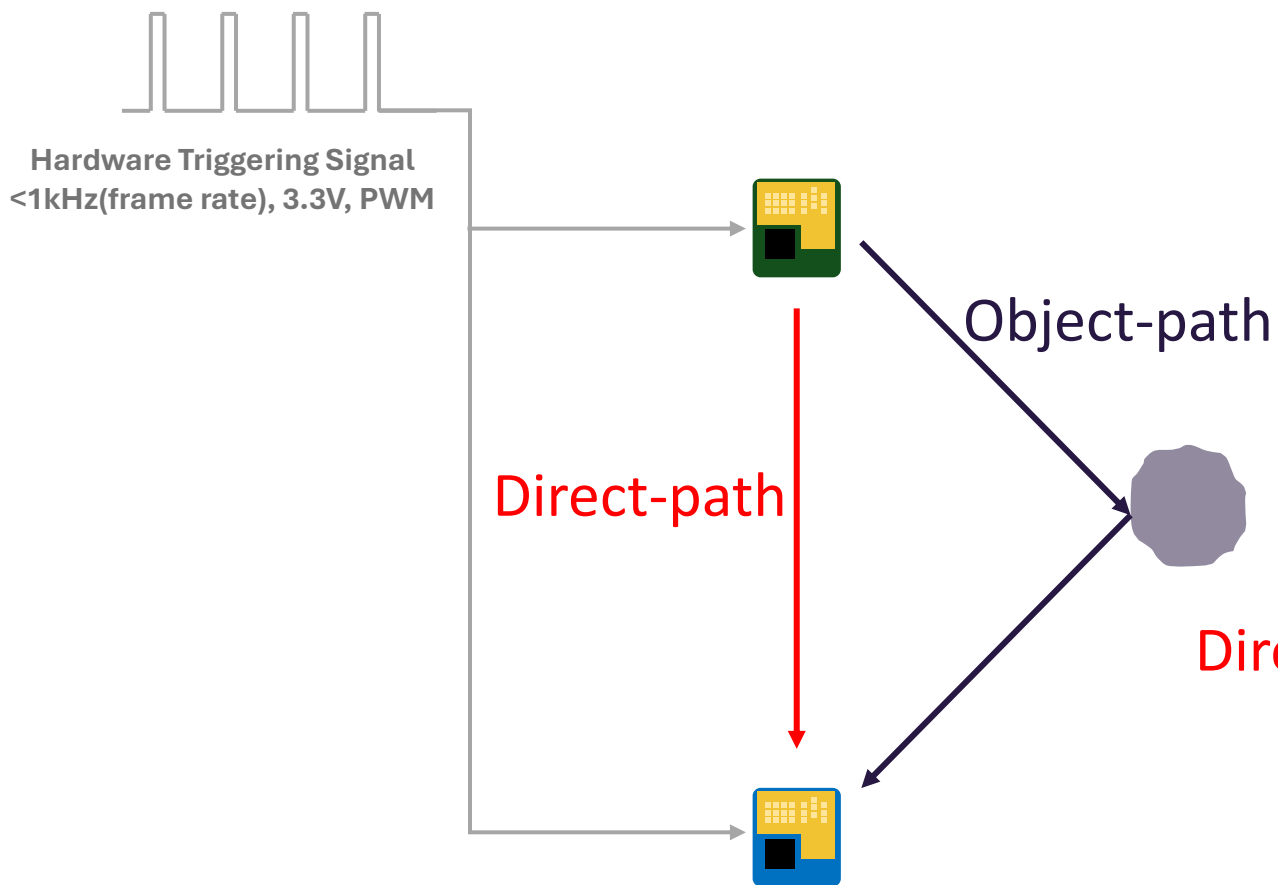
Distributed Setup: Spatial Channel Selection



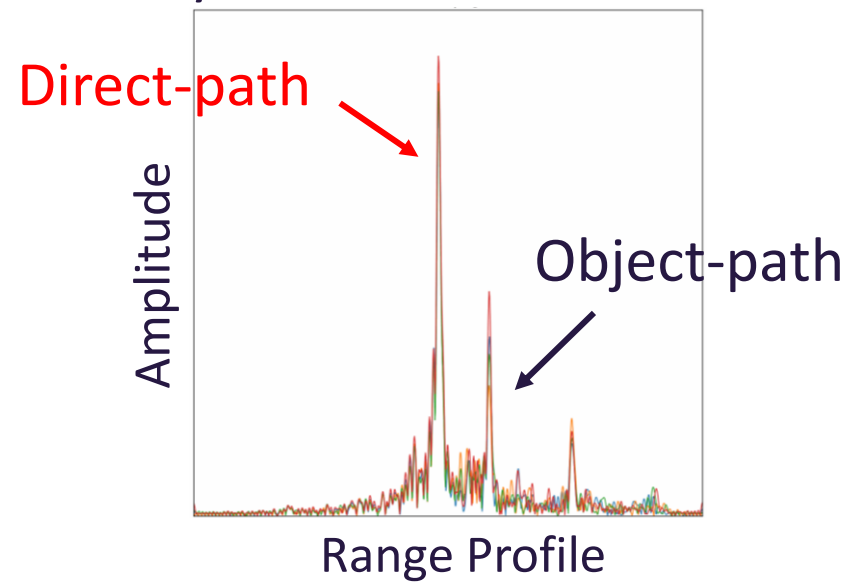


Synchronizing 77GHz FMCW Radars

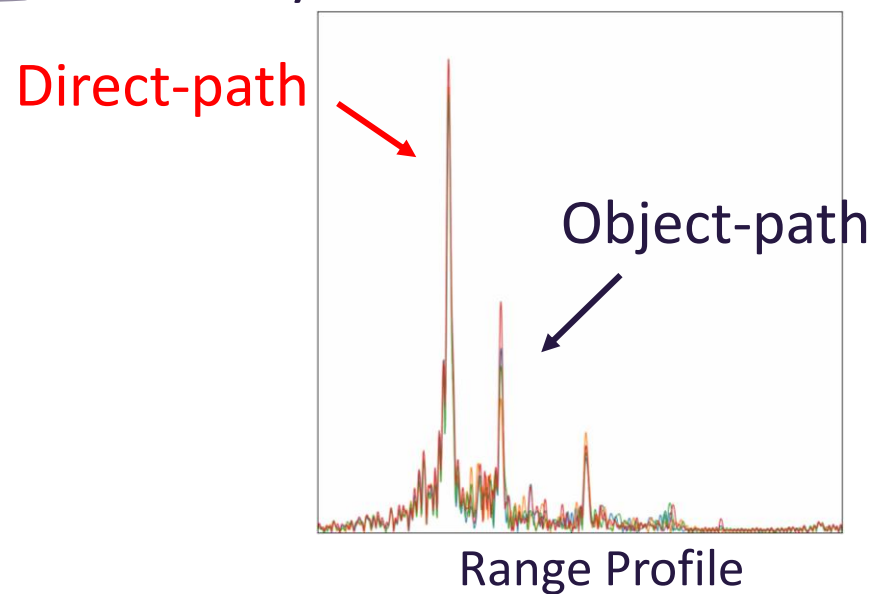




Unsynced Bi-static Channel



Synced Bi-static Channel



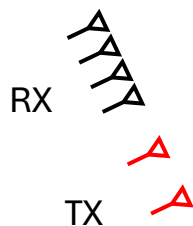
Multi-static Radar Array



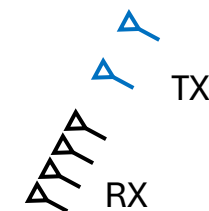
Multi-static Radar Array

Distributed radars are combined as one big multi-static radar

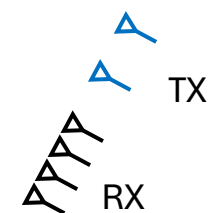
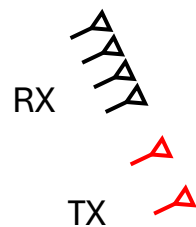
No concept of radars, but **antennas**



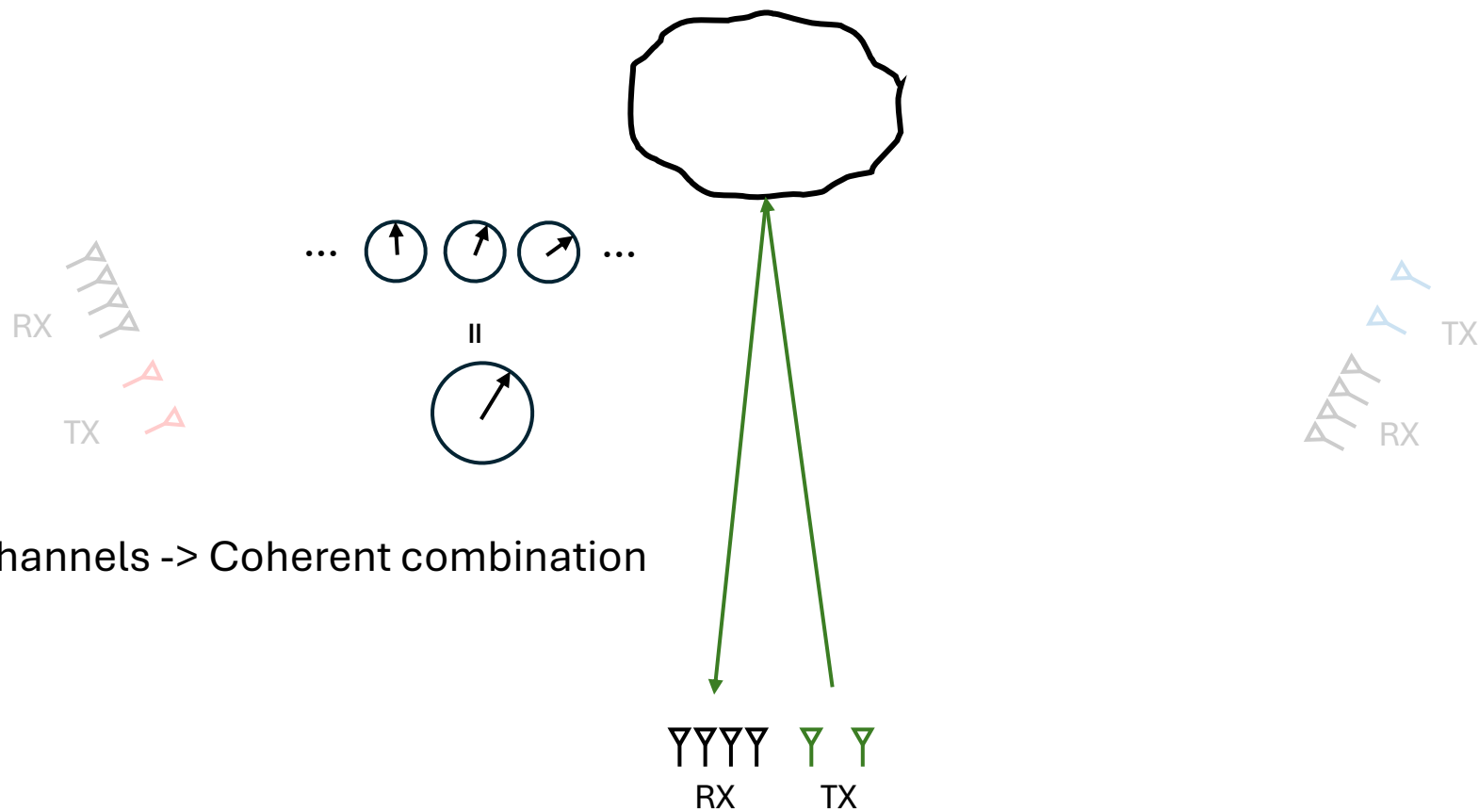
Sparse Multi-static Antenna Array



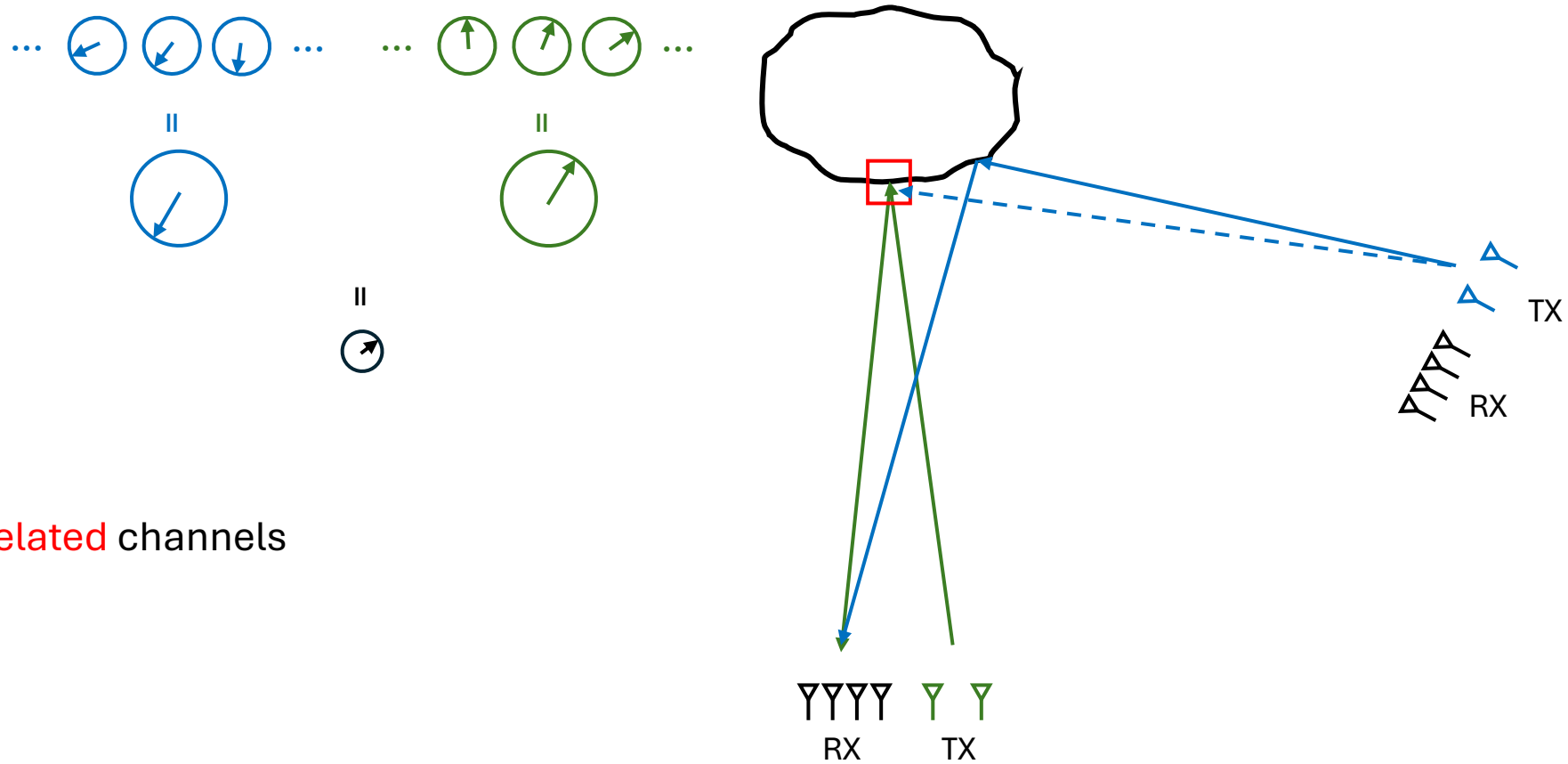
Sparse Multi-static Array Imaging



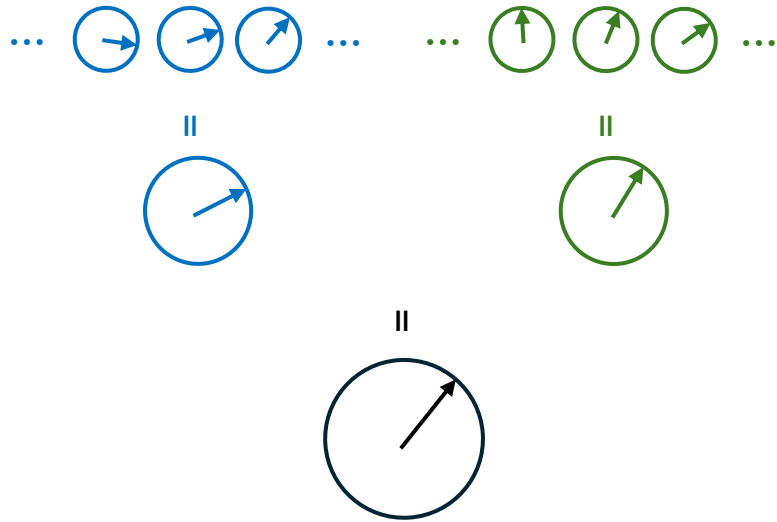
Multi-static Imaging: Back-projection



Correlated Channels and Uncorrelated Channels

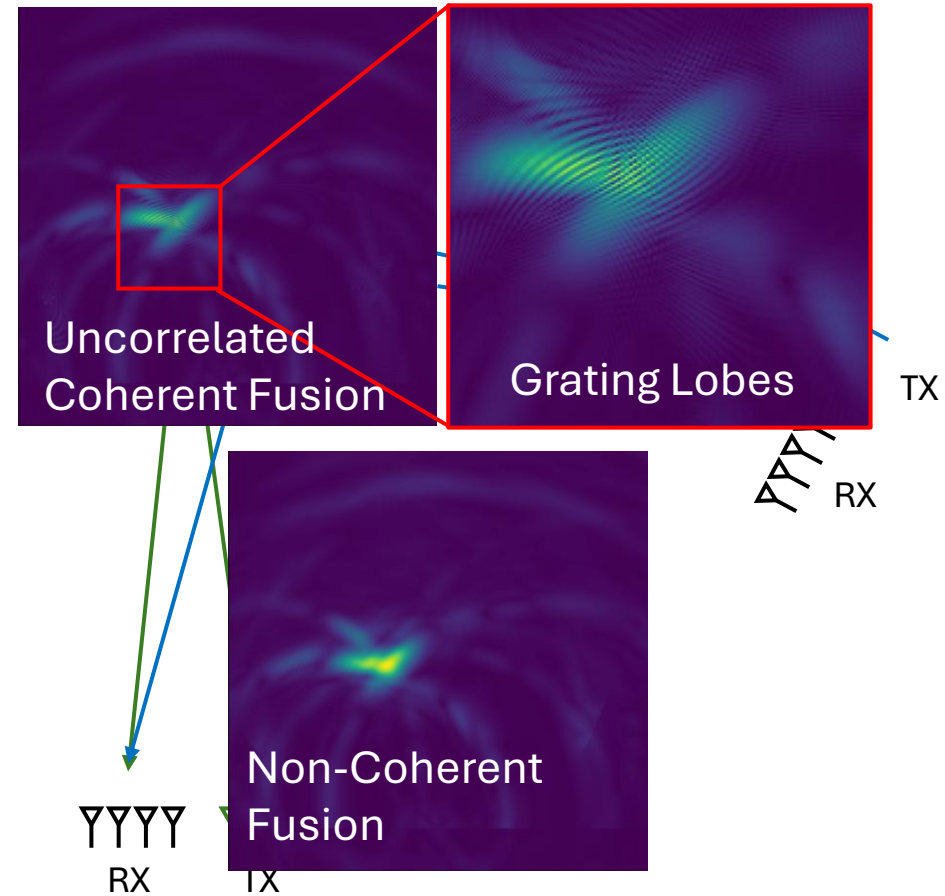


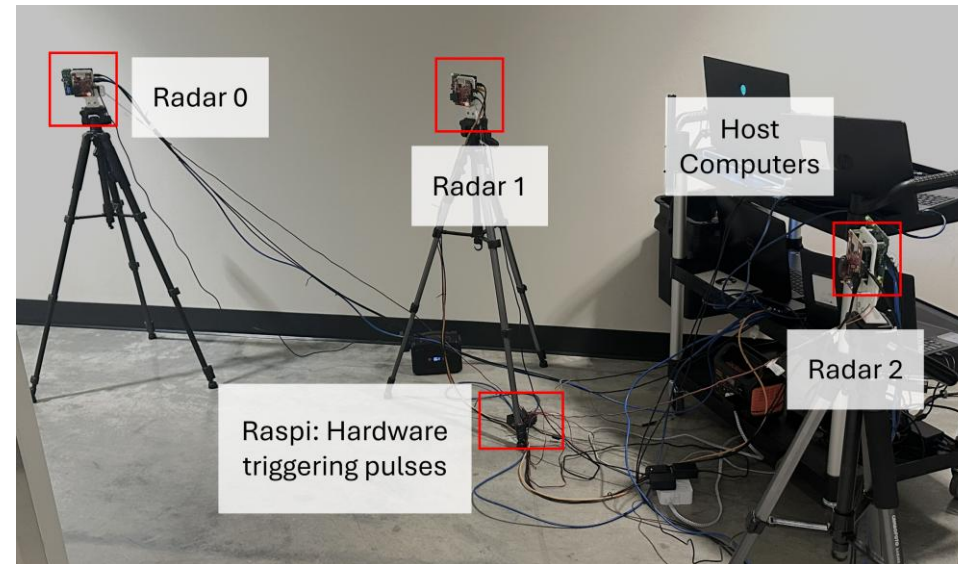
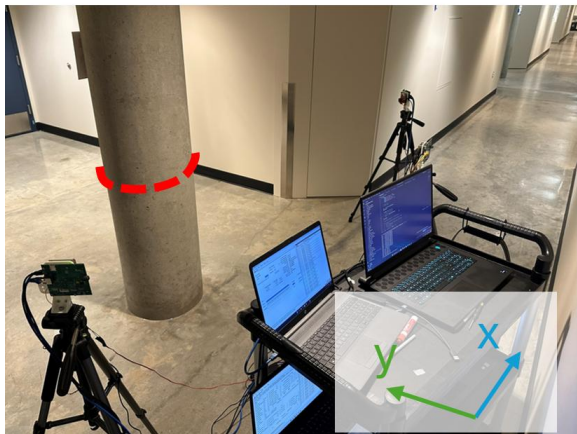
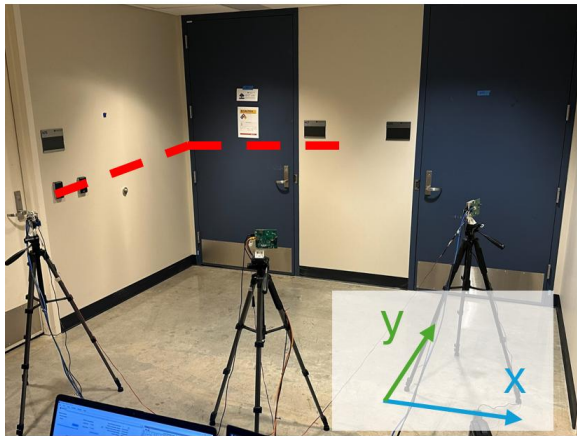
Correlated Channels and Uncorrelated Channels

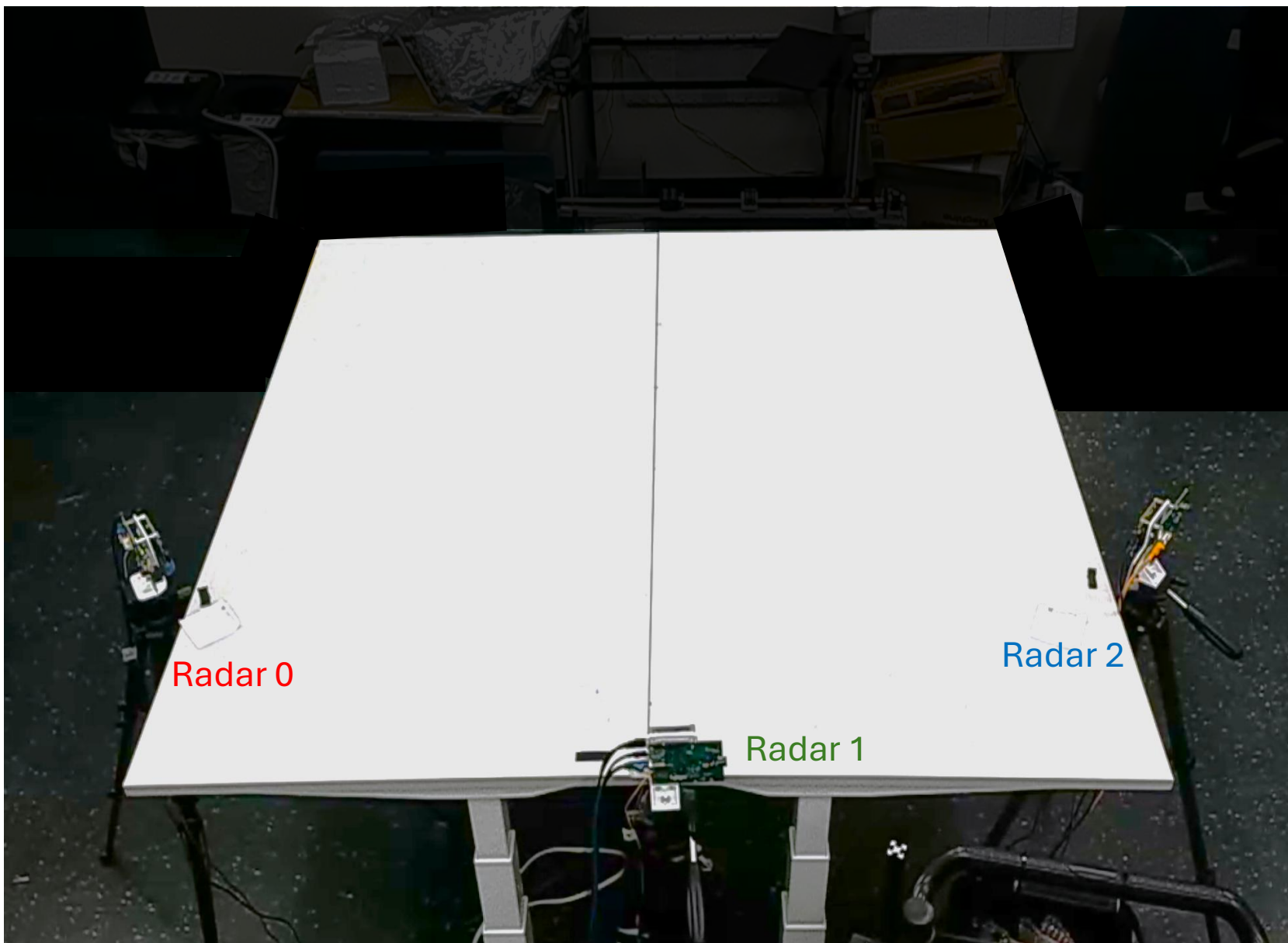


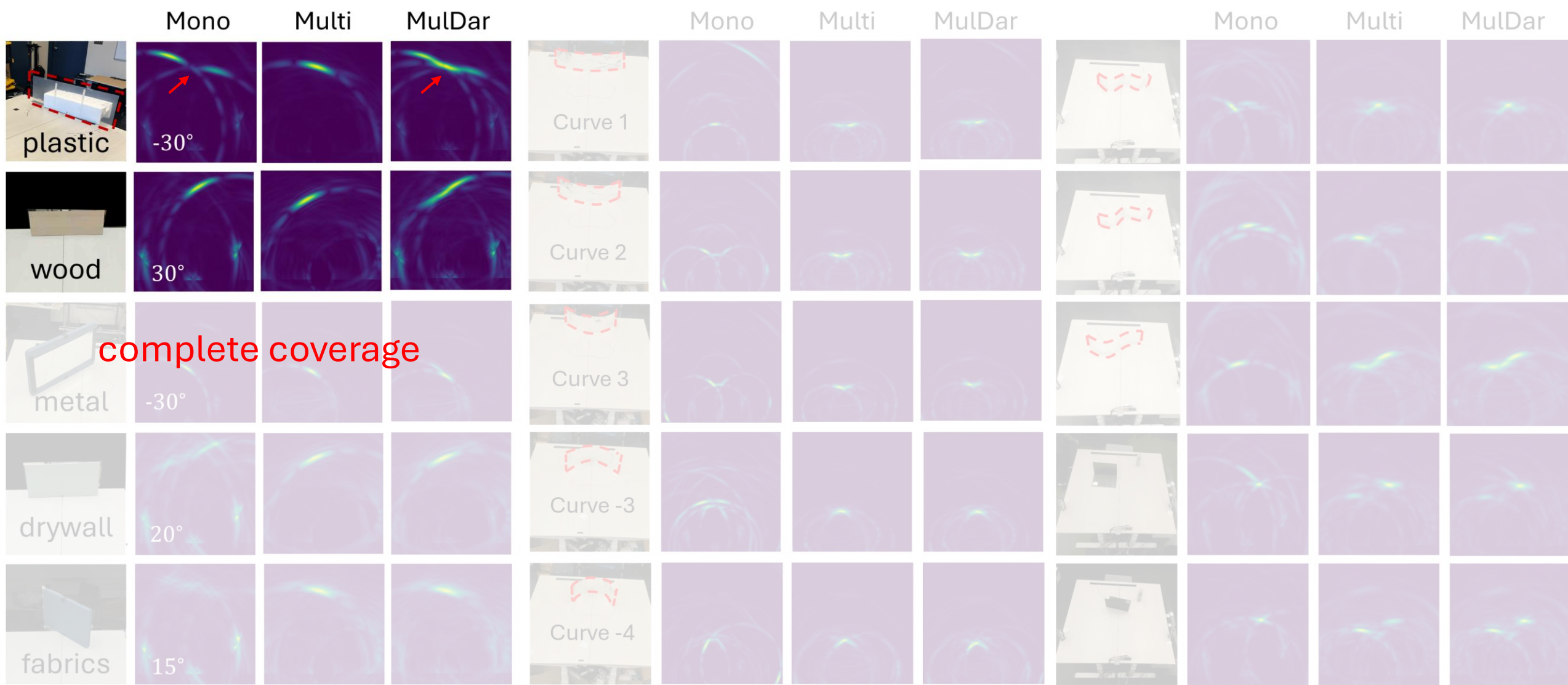
Uncorrelated channels

-> Coherent combination creates grating lobes





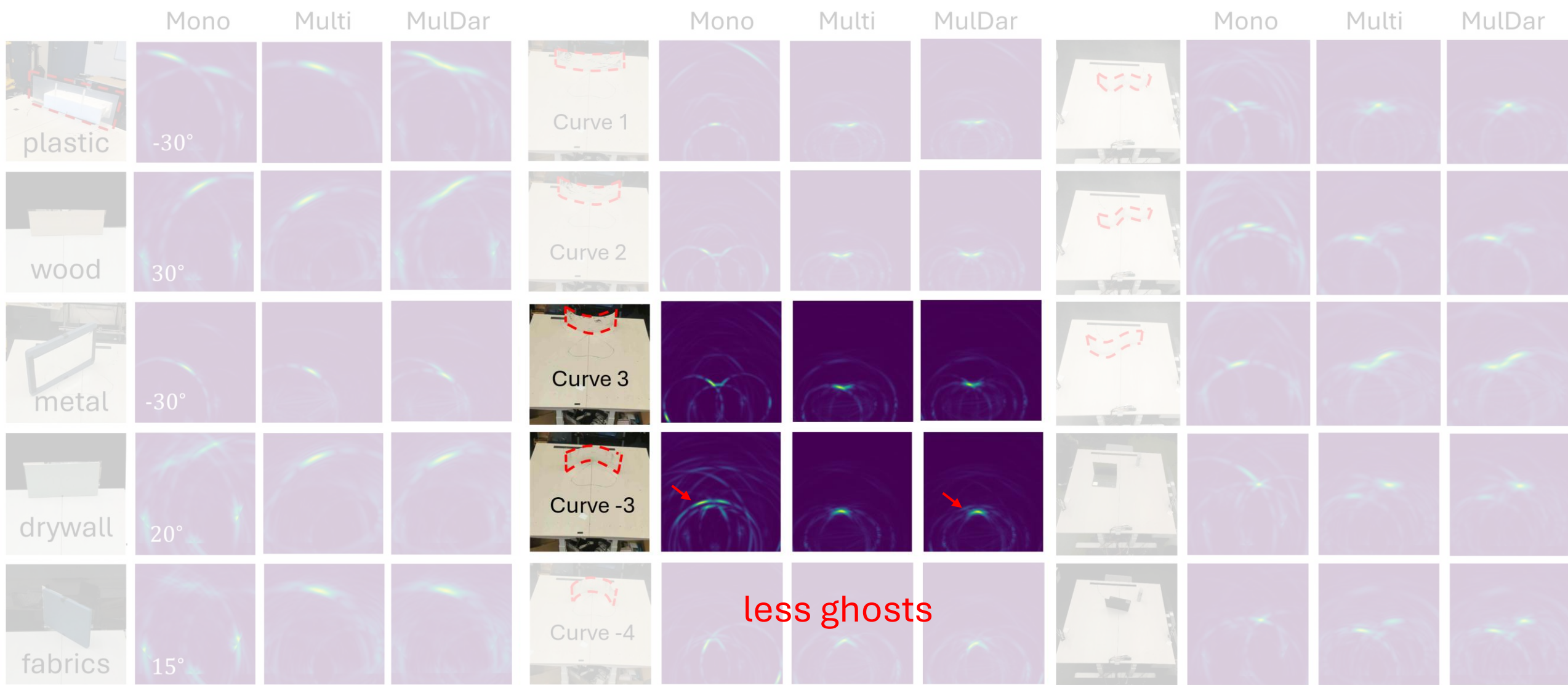




(a) Planar scenes

(b) Curved scenes

(c) Deformable & real scenes

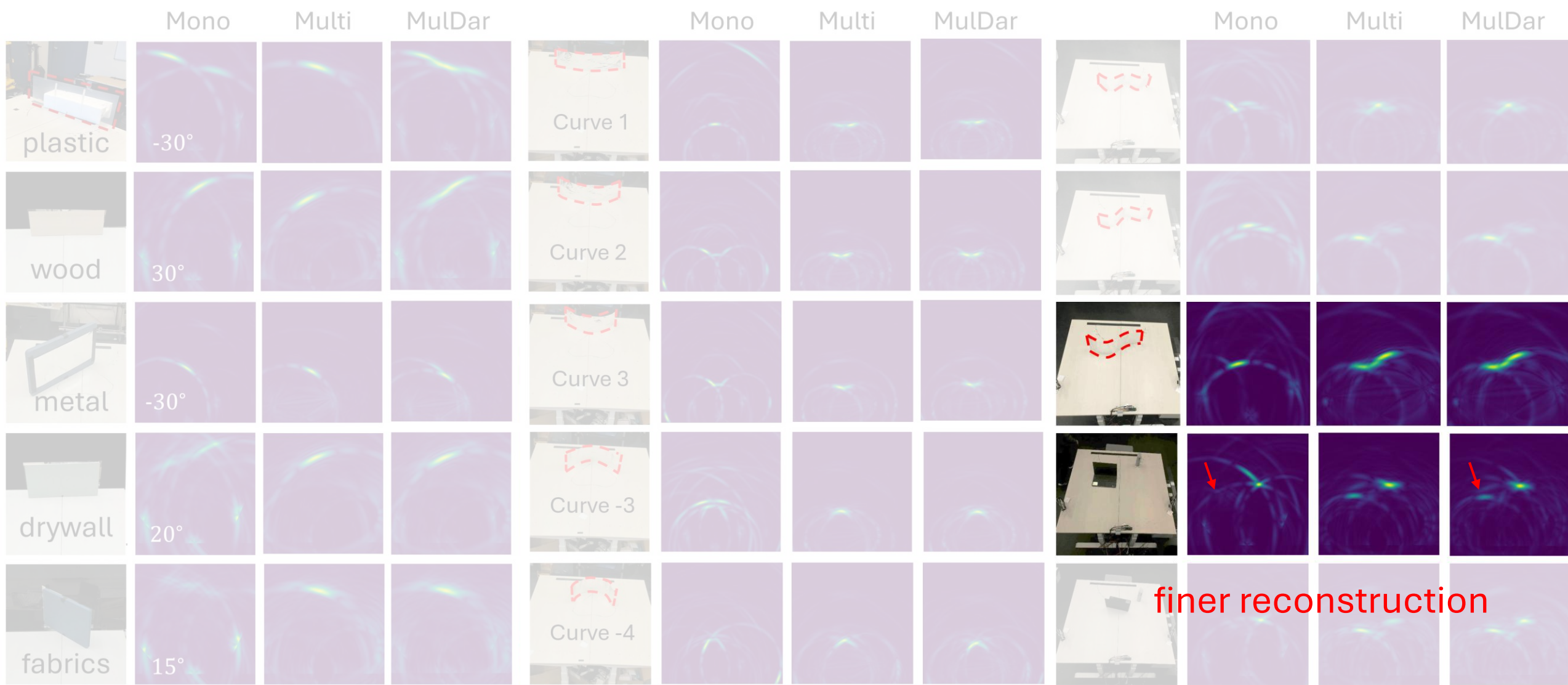


less ghosts

(a) Planar scenes

(b) Curved scenes

(c) Deformable & real scenes

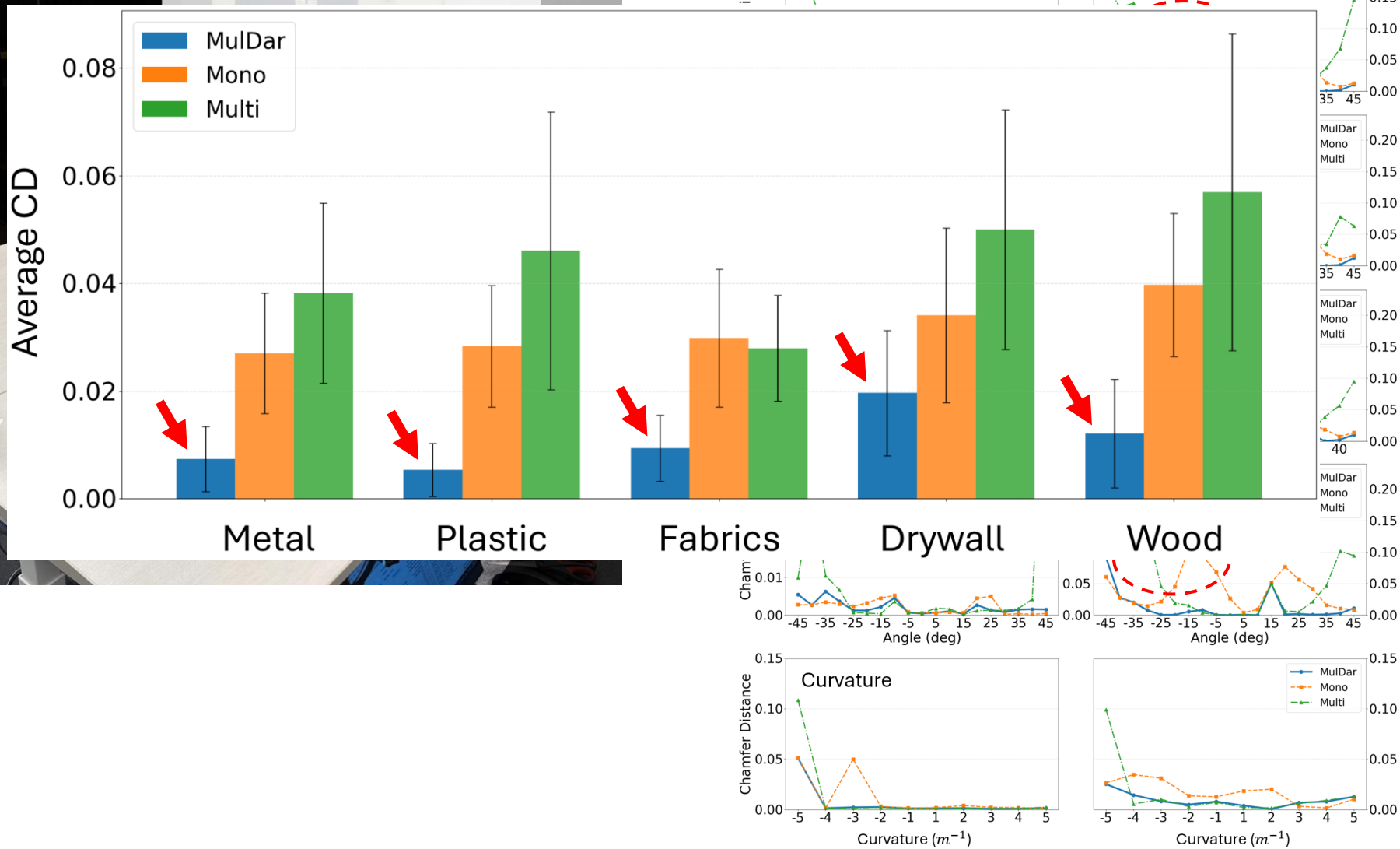
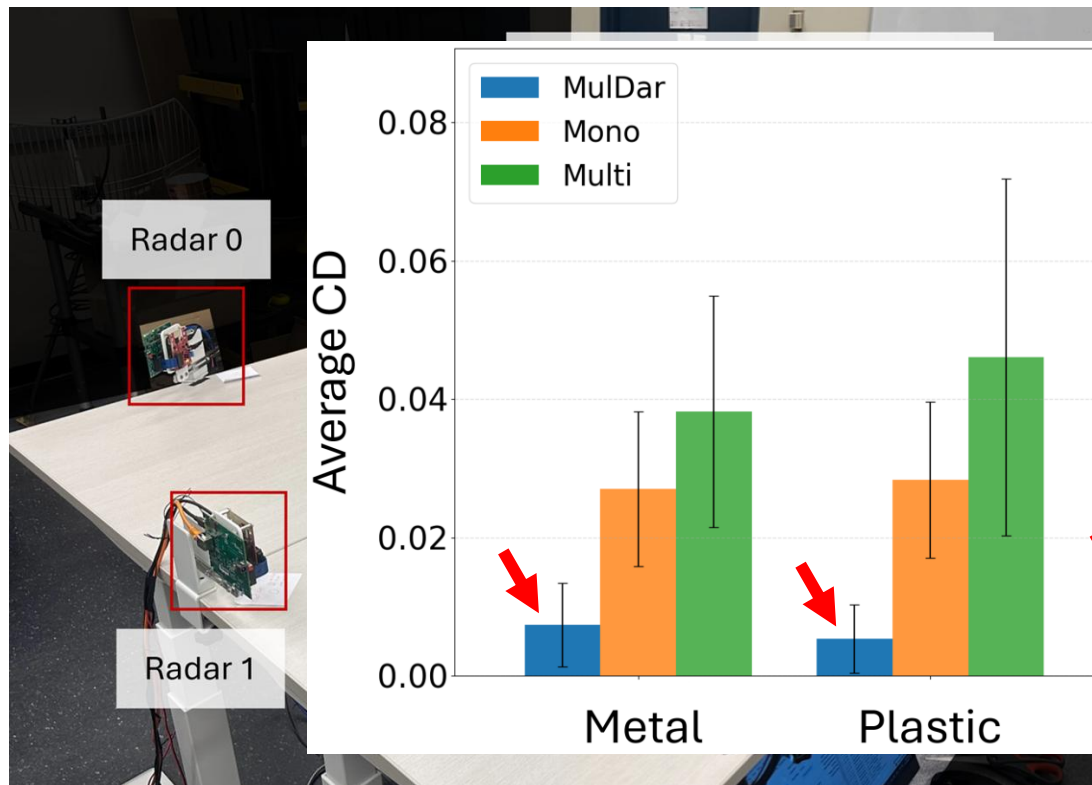


finer reconstruction

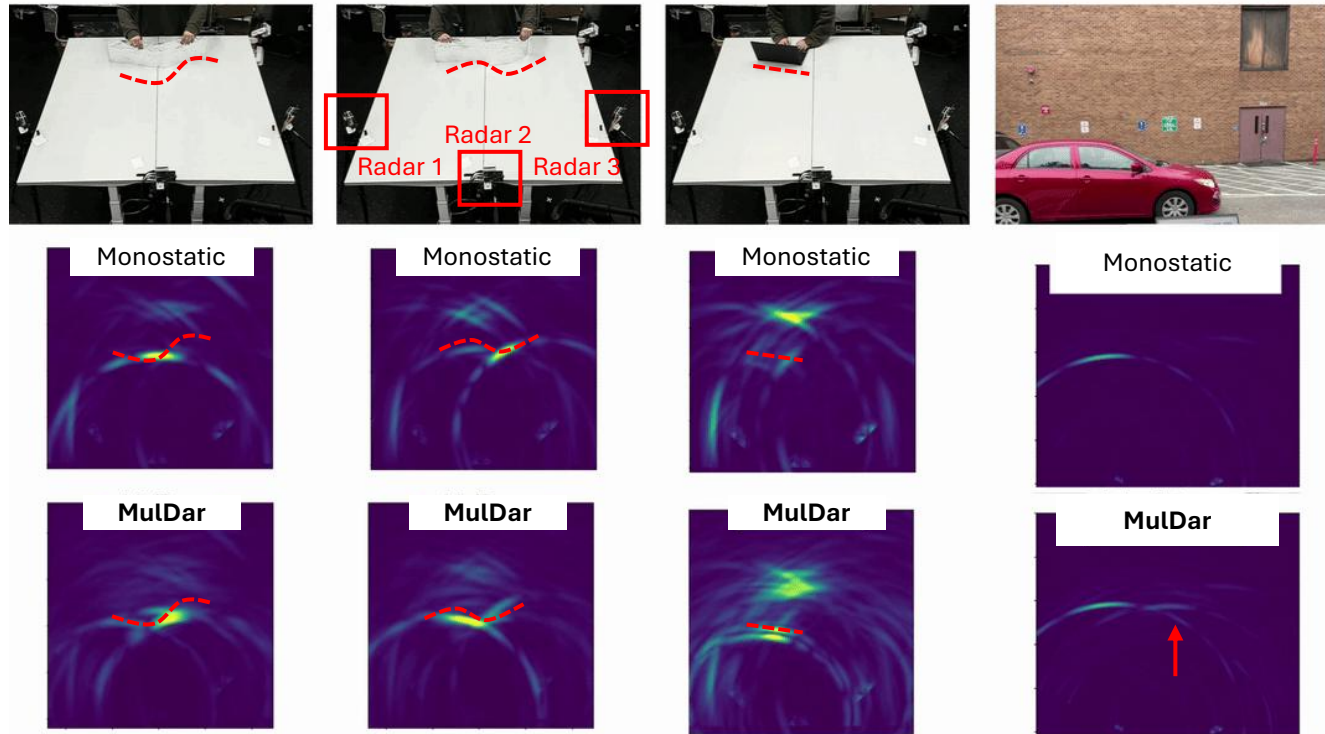
(a) Planar scenes

(b) Curved scenes

(c) Deformable & real scenes



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MulDar
Code & Dataset & Tutorials



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Questions?
Let's connect!



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